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Anal. Chem., **Just Accepted Manuscript** • Publication Date (Web): 22 May 2019

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Multifunctional Fe₃O₄@Polydopamine@DNA-fueled Molecular Machine for Magnetically Targeted Intracellular Zn²⁺ Imaging and Fluorescence/MRI Guided Photodynamic-Photothermal Therapy

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ABSTRACT: For the precise treatment of tumors, it is necessary to develop a theranostic nanoplatform that has both diagnostic and therapeutic functions. In this paper, we designed a new theranostic probe for fluorescence imaging of Zn²⁺ and fluorescence/MRI guided magnetically targeted photodynamic-photothermal therapy. The fluorescence imaging of Zn²⁺ was based on an endogenous ATP-driven DNA nanomachine that could perform repetitive stand displacement reaction. It modifies all units on a single PDA/Fe₃O₄ nanoparticle containing a hairpin-locked initiated strand activated by a target molecule in cells, a two-stranded fuel DNA triggered by ATP and a two-stranded DNA track responding to initiated strand and fuel DNA. After entering the cell, the intracellular target Zn²⁺ initiates the nanomachine *via* autocatalytic cleavage reaction, the machine programmatically and gradually runs on the assembled DNA track *via* fuel DNA driving and intramolecular toehold-mediated strand displacement reaction. The Fe₃O₄ core first exhibits magnetic targeting, increasing the ability of nanoparticles to enter tumor cells at the tumor site. The Fe₃O₄ could also be employed as powerful magnetic resonance imaging (MRI) contrast agent and guided therapy. Using 808 nm laser and 635 nm laser irradiation together at the tumor site, the PDA nanoshell produced excellent photothermal effect and the TMPyP4 molecules entering the cell generated reactive oxygen species, followed by cell damage. A series of reliable experiments suggested that the Fe₃O₄@PDA@DNA nanoprobe showed superior fluorescence specificity, MRI, remarkable photothermal/photodynamic therapy effect and favorable biocompatibility. This theranostic nanoplatform offered a split-new insight into tumor fluorescence and MRI diagnosis as well as effective tumor therapy.

As a crucial part of precision medicine, accurate diagnosis is of great significance for efficient tumor therapy^{1,2}. MicroRNAs³⁻⁶ and telomerase^{7,8} were usually employed as diagnostic biomarkers to detect tumor, because their abnormal expressions were associated with tumor occurrence and progression. Apart from the familiar cancer-indicated biomarkers, the disorder of metal ion homeostasis in biological systems is also a critical tumor-indicated signal^{9,10}. Zn²⁺ plays a pivotal role in various basic biological processes¹¹. It was obvious that the destruction of Zn²⁺ homeostasis has long been associated with prostate cancer and brain cancer¹². Recently, Kathryn M. Taylor *et al.* discovered that the protein kinase CK2 was a switch which transports free Zn²⁺ to breast cancer cells, suggesting that superabundant Zn²⁺ was one of the significant cancer-indicated biomarkers^{13,14}. In the past time, a variety of methods for the detection of Zn²⁺ have been reported. The traditional methods include atomic absorption spectrometry¹⁵, plasma mass spectrometry¹⁶, plasma atomic emission spectrometry¹⁷, electrochemical¹⁸ method and so on^{19,20}. However, due to its own limitations, traditional

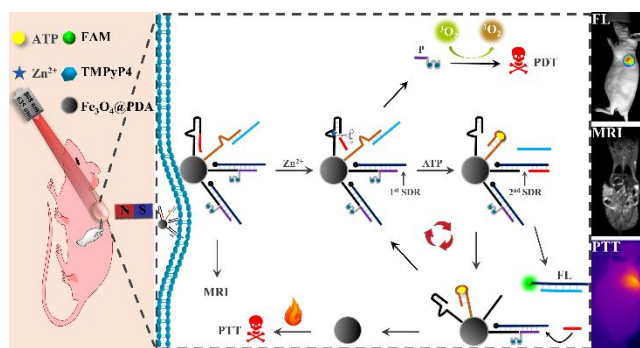
methods cannot monitor the metabolism of Zn²⁺ in living cells. Thanks to its high stability and convenience, fluorescence imaging is an effective means for *in situ* observation of intracellular Zn²⁺. Lu *et al.*²¹ successfully applied photothermal activation and DNAzyme to fluorescence imaging of intracellular Zn²⁺. Tan *et al.*²² designed a composite nanocapsule for fluorescence imaging of intracellular Zn²⁺ and other metal ions. As expected, most of the fluorescence detection methods have achieved the corresponding results in target monitoring, however, it is necessary to add external Zn²⁺ to the cells before imaging according to their lack of sensitivity. In many circumstances, the concentration of Zn²⁺ is lower than the concentration threshold that is required to effectively activate the imaging probe, which is a major technical obstacle to Zn²⁺-guided cancer diagnosis and treatment. Therefore, it is urgent to enhance the detection limit of fluorescence imaging for intracellular Zn²⁺. To achieve a more sensitive detection of Zn²⁺, the DNAzyme probe was prepared to achieve signal amplification⁹. Some signal amplification ideas have been designed for fluorescence

imaging in cells, for instance chain-substituted cascade amplification reactions, hairpin self-assembly reactions, etc²³⁻³¹.

Recently, DNA nanomachine have attracted widespread attention in the establishment of different DNA assembly structures and DNA biosensors³². The DNA molecular machine is a well-designed DNA system that employs DNA or other small molecules as a fuel to self-assemble using pre-simulated sequences. These DNA nanomachines are usually designed to work according to toehold-mediated strand displacement reaction (TSDR), in which a ssDNA sequence is hybridized to a partial dsDNA by a dangling region called toehold, replacing the original shorter complementary strand in a stepwise branch transfer^{33,34}. TSDR can establish a programmable DNA hybridization system by managing designated base sequence order and length of the toehold^{35,36}. In view of the entropy-triggered, oligonucleotide-driven and enzyme-free catalyzed characteristics of TSDR, DNA molecular machine was applied to construct different DNA biosensors to detect mRNA, proteins, and some metal ions^{37,38}. Although these nanomachines have made great progress in biological applications, the external synergistic delivery of fuels, addition of necessary supplemental elements or factor interference in external environment have made the design and handling procedures difficult³⁹. Besides, the unsynchronized delivery of each unit into cells will reduce the domination of the temporal and spatial distribution of fundamental units of DNA nanomachine and the assistant units in living cells. Exploiting a method for fluorescence imaging of endogenous Zn^{2+} based on an endogenous ATP-driven DNA nanomachine to perform repetitive strand displacement reaction and to realize Zn^{2+} guided photodynamic therapy (PDT) has great significance.

In PDT, photosensitizers generate lots of reactive oxygen species (ROS) to damage tumor cells under light irradiation⁴⁰⁻⁴². However, so far, PDT still has a lot of difficulties, such as specific targeting, biocompatibility and real-time PDT feedback etc^{43,44}. Extensive exploration in PDT has been made by modifying the targeting molecules on nanomaterials to improve tumor selectivity. However, this approach also faces limitations in the expression of receptors between patients. The use of external magnetic field to assimilate therapeutic drugs circulating in the bloodstream, thereby enhancing the enrichment of nanomaterials in the tumor region, is also an important part of the development of tumor targeting strategies^{45,46}. In many reports, magnetic nanoparticles have been exploited for photothermal therapy of tumors because of their strong absorbance in the near-infrared (NIR) light region and their property to convert light into heat⁴⁷⁻⁴⁹. However, magnetic Fe_3O_4 nanoparticles (NPs) are not excellent photothermal therapy (PTT) agent depending on their inefficient optical-thermal energy conversion efficiency and low amounts of accumulation on the unmodified surface of the target tissue, besides, the reticuloendothelial system can easily absorb nanoparticles, and they quickly remove blood circulation, which hinders the efficacy of PTT. In order to solve the problem of rapid clearance, various polymer decorated magnetic Fe_3O_4 NPs have been developed. Dopamine, a critical noradrenaline precursor substance naturally occurring in living organisms, could spontaneously self-polymerize without the need for additional oxidants under

alkaline conditions⁵⁰. After self-polymerization of dopamine, polydopamine (PDA) exhibits outstanding biocompatibility and biodegradability. PDA also has higher photothermal conversion efficiency (40%), which has been proven to be an ideal photothermal material for ablating of tumor cells without affecting healthy tissues⁵¹. Furthermore, PDA has been diffusely employed as a quencher in the biological field, especially in combination with DNA probes. By further covering the Fe_3O_4 core with PDA, it is possible to prepare a multifunctional composite nanodevice and enhance its photothermal performance, surpassing the effect of using only Fe_3O_4 core or PDA sphere⁵². As an excellent magnetic resonance imaging (MRI) agents, Fe_3O_4 NPs has been broadly applied for tumor diagnosis⁵³. Owing to excellent physicochemical characteristics of Fe_3O_4 NPs and PDA, the exploitation of nucleic acid probe loading PDA- Fe_3O_4 NPs-based theranostic system by a facile assembly procedure is urgent to prepare for sensitive monitoring of Zn^{2+} and photodynamic-photothermal therapy.



Scheme 1. Multifunctional Fe_3O_4 @Polydopamine@DNA-fueled molecular machine for intracellular Zn^{2+} imaging and fluorescence/MRI guided magnetically targeted photodynamic-photothermal therapy.

As shown in **Scheme 1**, in the presence of intracellular Zn^{2+} , the hairpin DNA containing substrate strands and DNAzyme strands was cleaved by itself to generate triggered DNA (T), which could initiate first TSDR and release photosensitizer TMPyP4 (P). Then, sufficient dissociative ATP in cells involuntarily binds to the ATP aptamer, leading to the release of fuel DNA, accompanied by the initiation of the second TSDR. The fuel DNA was essential to realize the recycle of two cascaded TSDRs. Each recycled reaction would use one fuel DNA molecule to liberate the initiator DNA for next recycled reaction with concurrent fluorescence restoration of FAM in linker DNA with the help of DNA hybridization. The cascade of TSDR to produce abundant hybrid linker DNA/fuel DNA, which led to an increased fluorescence response. At the same time, the Fe_3O_4 provided powerful MRI contrast agent. Furthermore, using near-infrared light and 635 nm-red laser irradiation jointly at the tumor region, a photothermal effect produced by PDA and singlet oxygen (1O_2) triggered by P were produced in the living cells, thereby causing cell death. As carrying out comprehensive experiments, the superior performance of the nanoplatfrom confirms its application prospects in tumor.

EXPERIMENTAL SECTION

Detection of Zn^{2+} in Vitro. The functionalized $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe was added to 1 mL PBS solution to reach a concentration of $100 \mu\text{g mL}^{-1}$. Various concentrations of Zn^{2+} were respectively dumped into 1 mL $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ solution ($60 \mu\text{g mL}^{-1}$) containing 10 mM ATP and reacted in constant temperature shaker at 37°C for 3 h, the supernatant was collected by centrifugation at 10000 rpm for 8 min. Subsequently, the fluorescence intensity of the suspension was detected by a fluorescence spectrophotometer.

Intracellular Imaging of Zn^{2+} . MCF-7 cells grow on 35-mm confocal dishes. Before ingesting nanoprobe, the cells were incubated with TPEN (a membrane-permeable Zn^{2+} chelating agent that possesses the ability of effectively removing Zn^{2+}), DTDP (a reagent that accelerates Zn^{2+} to get rid of Zn^{2+} -binding proteins) or $5 \mu\text{M}$ Zn^{2+} for 30 min, respectively. Afterward, the $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe ($60 \mu\text{g mL}^{-1}$) was delivered into MCF-7 cells in culture medium for 4 h in a magnetic field, and then the cells were cleaned with PBS. Finally, fluorescence images of cells were visualized under an Olympus FV10i confocal scanning system.

In Vivo Fluorescence and MR Imaging. For *in vivo* fluorescence image or MR imaging, the tumors were intratumorally injected with $50 \mu\text{L}$ of $200 \mu\text{g mL}^{-1}$ $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe dispersion and then incubated under the guidance of magnetic field. Fluorescence images of mice in different time groups was visualized using a small animal imager. For *in vivo* MR imaging, after incubated 6 h, the mice were anaesthetized and imaged with a special animal magnetic coil.

In Vivo Synergistic PTT and PDT. Before the experiment, four groups of MCF-7 tumor-bearing nude mice were prepared at random ($n = 5$). Four groups of tumors were intratumorally injected with $50 \mu\text{L}$ of $200 \mu\text{g mL}^{-1}$ $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe dispersion and incubated for 6 h in a magnetic field. For PTT, tumors were exposed to 808 nm laser (0.8 W cm^{-2}) for 10 min. For PDT, tumors were exposed to 635 nm laser (0.5 W cm^{-2}) for 5 min. For the combination of PTT and PDT, tumors were irradiated by 808 nm laser (0.8 W cm^{-2}) for 10 min, followed by 635 nm laser (0.5 W cm^{-2}) for 5 min. For control group, tumors do not do any light treatment. The synergistic treatment was performed on 0th day and 8th day.

RESULTS AND DISCUSSION

Characterization of $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ Nanoprobes. The procedure to synthesize the $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs was based on a modified emulsion polymerization protocol⁵⁴. The polystyrene-trapped magnetic $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs were wrapped by PDA nanoshell under alkaline conditions ($\text{pH} = 8.5$). Transmission electron microscopy (TEM) images displayed that magnetic Fe_3O_4 NPs were clearly distributed on the polystyrene nanobead with diameter of about $85 \pm 5 \text{ nm}$ (**Figure 1A**). The diameter of $\text{Fe}_3\text{O}_4@\text{PDA}$ was approximately $115 \pm 5 \text{ nm}$, and thus the thickness of the visible PDA nanoshell was roughly 20 nm (**Figure 1B**). The scanning electron microscope (SEM) further verified the size and morphology of the Fe_3O_4 NPs and $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs (**Figure 1E and 1F**). The dynamic light scattering (DLS) data showed that the average size of Fe_3O_4 NPs was 97 nm and $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs was 131 nm (**Figure 1C**). The particle size measured by DLS was much larger than by TEM due to the

hydration of nanoparticles and the potential for micro-polymerization. In general, $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs have relatively narrow size distribution and most of its dimensions are within the biosafety range (below 150 nm). To examine molecular composition and structure of the final product, polystyrene-trapped Fe_3O_4 NPs and $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs were analyzed using FTIR spectra (**Figure 1D**). For polystyrene-trapped Fe_3O_4 NPs, polystyrene showed two intense absorption band at 696 cm^{-1} and 754 cm^{-1} which were caused by the C-H bond bending vibration of monosubstituted benzene. In addition, two intense absorption bands at 1620 cm^{-1} and 584 cm^{-1} were observed due to the tensile vibration of Fe-O and H-O-H groups. After the dopamine coating was applied, $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs displayed a strong absorption band from 1800 cm^{-1} to 1000 cm^{-1} , which was attributed to the tension of C-O in phenol compounds and the tensile vibration of aromatic rings. X-ray photoelectron spectroscopy (XPS) was introduced to verify the components and structure. **Figure 1G and S1** showed that $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs contained four elements (C, N, O and F) corresponding to the characteristic peaks of C 1s, N 1s, O 1s and Fe 2p. The crystal structure of polystyrene-trapped Fe_3O_4 NPs and $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs were determined by X-ray diffraction (XRD) analysis as displayed in **Figure 1H**. The main characteristic diffraction peaks of 30.70° , 36.06° , 43.76° , 54° , 57.82° and 63.46° were attributed to the planes of Fe_3O_4 , which were consistent with magnetic iron oxide nanoparticles in the JCPDS card No. 19-0629. It is worth mentioning that, $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs and Fe_3O_4 NPs showed nearly identical diffraction peaks. The result indicated that the $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs phase of Fe_3O_4 was efficiently formed during synthetic process, and dopamine and polystyrene had no effect on it.

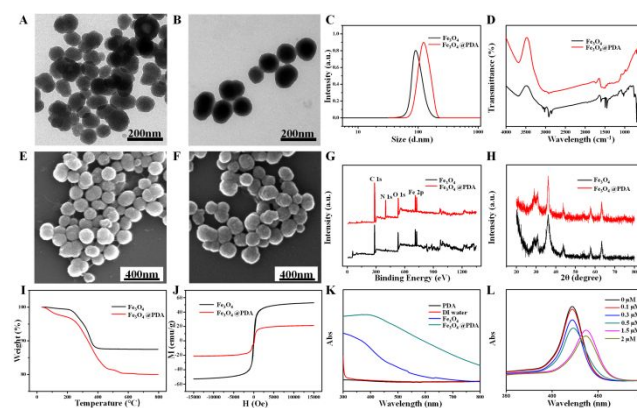


Figure 1. TEM images of (A) Fe_3O_4 NPs, (B) $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs. (C) DLS size distributions of Fe_3O_4 NPs and $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs. (D) The FTIR spectrum of Fe_3O_4 NPs and $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs. SEM images of (E) Fe_3O_4 NPs, (F) $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs. (G) Survey XPS of Fe_3O_4 NPs and $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs. (H) XRD patterns of Fe_3O_4 NPs and $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs. (I) TGA curves of Fe_3O_4 NPs and $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs. (J) Magnetization curves of Fe_3O_4 NPs and $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs. UV-vis absorption spectra of (K) Fe_3O_4 NPs, $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs and (L) P with aptamer of various concentration (0-2 μM).

The coating efficiency of dopamine was quantitatively analyzed by thermogravimetry (TGA), the weight percentage of polystyrene-trapped Fe_3O_4 NPs and $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs were 12.54% and 19.96%, respectively (**Figure 1I**). The raise in weight percentage was attributed to the presence of PDA nanoshell. According to the results of analysis and calculation, the coating amount of the PDA was 7.42%. To verify the

magnetic characteristics, Fe_3O_4 NPs and $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs were measured by cyclic magnetic field between -15 k and $+15\text{ k}$ Oe at 300 K . After the dopamine coating was applied, the saturation magnetization dropped from 52.51 emu g^{-1} (Fe_3O_4 NPs) to 21.66 emu g^{-1} ($\text{Fe}_3\text{O}_4@\text{PDA}$ NPs) and no exhibit any evident remanence or coercivity (**Figure 1J** and **S4A**). The results indicated that the superparamagnetic property of $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs has not been affected by dopamine coating. Although the saturation magnetizations of $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs dropped, it was enough to play the role of magnetic targeting.

The UV-vis absorption spectrum revealed the variations during the synthesis of $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs (**Figure 1K**). After covering with PDA, the absorption peak of Fe_3O_4 NPs broadens and red shifts, which may be attributed to the accumulation of PDA on Fe_3O_4 NPs, indicating that its potential as a PTT reagent was significant (**Figure S2**). **Figure 1L** shows that the UV-vis absorption spectra of P exhibited the absorption peak at 422 nm . By adding functionalized AS1411 aptamer, the absorption peak of P was reduced and the peak site was shifted to 435 nm . The hyperchromicity of the Soret band confirms the physical adsorption between P and G-quadruplex of AS1411 DNA. In addition, once the P molecules was completely assembled on the AS1411 DNA, even if more AS1411 cDNA was added, there would be no significant change in the Soret band.

Feasibility of the Strategy for Zn^{2+} Detection. It has been reported that the sufficient quinone groups in PDA nanoshell could undergo spontaneous Michael addition and Schiff base reactions with thiol groups to densely conjugate functionalized DNA⁵⁵. In addition, PDA nanoshell was also an efficient quencher. Since FAM on functionalized DNA was close to PDA nanoshell, the FAM fluorescence was in the “turned-off” state because FAM was quenched by a fluorescence resonance energy transfer (FRET) mechanism. As shown in **Figure 2A**, the complete DNA functionalized $\text{Fe}_3\text{O}_4@\text{PDA}$ nanoprobe exhibit very weak fluorescence, demonstrating the quencher effect of PDA (line c). On the other hand, free thiol DNA functionalized $\text{Fe}_3\text{O}_4@\text{PDA}$ nanoprobe exhibit fluorescence intensity almost identical to FAM-DNA alone (line a and b), indicating distance of FAM and PDA nanoshell was critical to the occurrence of FRET. Compared to control group (**Figure 2B**, line e), the DNA functionalized $\text{Fe}_3\text{O}_4@\text{PDA}$ nanoprobe in PBS containing 40 nM Zn^{2+} and sufficient ATP exhibited excellent fluorescent intensity (line b). Although the fluorescence efficiency of DNA functionalized $\text{Fe}_3\text{O}_4@\text{PDA}$ nanoprobe dispersion has a certain decrease compared with DNA probe dispersion in absence of $\text{Fe}_3\text{O}_4@\text{PDA}$ (line a), it does not affect the specificity and sensitivity of $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe. Expectedly, the reaction solution of DNA functionalized $\text{Fe}_3\text{O}_4@\text{PDA}$ nanoprobe has extremely weak fluorescence intensity in absence of Zn^{2+} (line c) or ATP (line d). The results confirmed that the initiation of Zn^{2+} and the drive of ATP were all prerequisites for the implementation of the nanomachine amplification.

The feasibility of the nanomachine amplification for Zn^{2+} detection was further evaluated by agarose electrophoresis. As shown in **Figure 2C**, the control solution exhibited three clear bands attributing to the hybrid strands of ATP aptamer and fuel DNA (part A), the annealed Zn^{2+} -DNAzyme strands (part

B) and the hybrid strands of SH-DNA, linker DNA and AS1411 aptamer (part C) (lane a). It was worth noting that the base numbers of part A and part B were almost the same, so a bright trailing band appears. Three clear bands verified that three-part hybrid DNA strands in absence of $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs were independent of each other and do not affect each other. In the presence of only Zn^{2+} (lane b), the supernatant of the reaction product shows a weak brightness band (below band) and a much brighter band (upper band), which should be attributed to the Zn^{2+} -DNAzyme cleavage reaction releasing T-DNA and first TSDR releasing AS1411 aptamer. In the presence of only ATP (lane c), only a much brighter band was observed, confirming that ATP binds to ATP aptamer, and further releasing fuel DNA. In the presence of Zn^{2+} and ATP (lane d), a much brighter band (upper band) with low mobility and two weak brightness bands (below band) were observed, indicating that the nanomachine amplification of Zn^{2+} detection was successfully implemented by two cascaded TSDRs. In addition, lane e indicates that the DNA functionalized $\text{Fe}_3\text{O}_4@\text{PDA}$ nanoprobe dispersion does not react without Zn^{2+} and ATP.

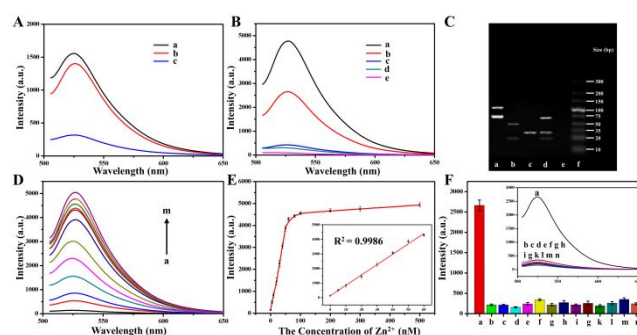


Figure 2. (A) Fluorescence spectra of only complete DNA nanoprobe (a), free thiol DNA functionalized $\text{Fe}_3\text{O}_4@\text{PDA}$ nanoprobe (b), complete DNA functionalized $\text{Fe}_3\text{O}_4@\text{PDA}$ nanoprobe (c). (B) Fluorescence spectra of only complete DNA nanoprobe with ATP and Zn^{2+} (a), complete DNA functionalized $\text{Fe}_3\text{O}_4@\text{PDA}$ nanoprobe with ATP and Zn^{2+} (b), with Zn^{2+} (c), with ATP (d), without ATP and Zn^{2+} (e). (C) Agarose gel electrophoresis image of only complete DNA nanoprobe (a), complete DNA functionalized $\text{Fe}_3\text{O}_4@\text{PDA}$ nanoprobe with Zn^{2+} (b), with ATP (c), with ATP and Zn^{2+} (d), without ATP and Zn^{2+} (e). (D) Fluorescence spectra of DNA functionalized $\text{Fe}_3\text{O}_4@\text{PDA}$ nanoprobe in the presence of ATP and various concentrations of Zn^{2+} (a to m: $0\text{--}500\text{ nM}$). (E) Fluorescence calibration curves for different concentrations of Zn^{2+} . Inset: standard linear calibration curves ($0\text{--}60\text{ nM}$). (F) Specificity of $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe over (a) Zn^{2+} , (b) Ca^{2+} , (c) Cu^{2+} , (d) Mg^{2+} , (e) Mn^{2+} , (f) Fe^{3+} , (g) K^{+} , (h) ascorbic acid, (i) dopamine hydrochloride, (j) glucose, (k) lysozyme, (l) mRNA, (m) telomerase, (n) uric acid.

The ability of the $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe to quantify Zn^{2+} was investigated. After mixing enough ATP, different concentrations of Zn^{2+} were added to the $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe dispersion to measure the fluorescence intensity. **Figure 2D** and **2E** displayed that the fluorescence intensity of the supernatant of $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe dispersion enhances with the concentration of Zn^{2+} increases ($0\text{--}500\text{ nM}$). The results showed a positive correlation with the Zn^{2+} concentration in the fluorescence spectrum, with a good linear range from 0.05 nM to 60 nM . The correlation linear equations for Zn^{2+} was $y = 70.9x + 141.7$, with coefficient $R^2 = 0.9986$, and the

detection limits of the nanoprobe was 0.05 nM. The detection range of the nanoprobe satisfies the response requirement of Zn^{2+} in living cell (0–300 nM). To evaluate the selectivity of the $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe to detect Zn^{2+} , other similar metal ions and small biomolecules were respectively applied to verify this strategy. As displayed in **Figure 2F**, the fluorescence intensity of Zn^{2+} (40 nM) was significantly stronger than that of other high concentrations of interferences. The excellent selectivity shows that such a $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ probe could detect Zn^{2+} in a multi-interfering biological environment due to the high specificity of Zn^{2+} -DNAzyme for Zn^{2+} .

Application of the $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ Nanoprobes in Imaging of Cellular Zn^{2+} . To investigate the generality of the strategy, cellular uptake and Zn^{2+} imaging of the $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe were observed using MCF-7 cells as a biological model. MCF-7 cells were first treated with 5 μM Zn^{2+} for 30 min to achieve enough metal ions uptake. As displayed in **Figure 3A**, MCF-7 cells treated with complete $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe and a magnetic field (ON state) exhibited a more pronounced green fluorescence image than complete $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe-treated MCF-7 cells without a magnetic field (OFF state), thus verifying this internalization of the magnetic guidance effect. Flow cytometry results further indicated that the magnetic guidance could enhance the endocytosis of $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe into tumor cells and exhibited much higher fluorescence signals (**Figure 3B**). In this designed strategy, fuel DNA loading on $\text{Fe}_3\text{O}_4@\text{PDA}$ via ATP aptamer plays a crucial role in the intramolecular toehold-mediated strand migration reaction. As expected, the fluorescence images showed no fluorescence in the absence of fuel DNA. For control group, endogenous Zn^{2+} imaging was investigated to test the feasibility of the $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe in untreated cells, since the concentration of Zn^{2+} in living cells was relatively low. Confocal fluorescence image results indicated that the control group was observed fewer and weaker fluorescence compared with other groups. These findings clearly proved that the intramolecular toehold-mediated strand migration reaction had the promise in affording higher amplification efficiency and better sensitivity in living cell imaging.

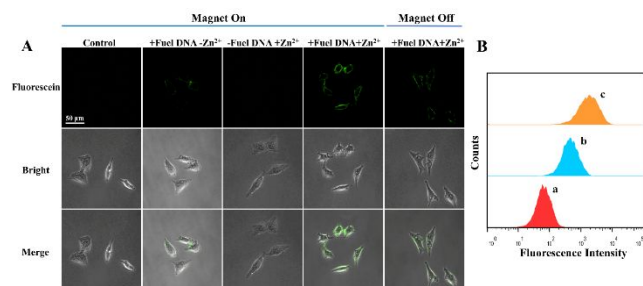


Figure 3. (A) Fluorescence images of MCF-7 cells incubated with DNA functionalized $\text{Fe}_3\text{O}_4@\text{PDA}$ nanoprobe under the guidance of magnetic field or not. (B) Flow cytometry assay of complete $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe without Zn^{2+} and magnetic field (a), in the presence of Zn^{2+} and without magnetic field (b), in the presence of Zn^{2+} and with magnetic field (c).

To further confirm that the cellular fluorescence imaging was generated by endogenous Zn^{2+} , we applied the proposed strategy to evaluate intracellular Zn^{2+} changes upon

stimulation of several drugs. As displayed in **Figure 4**, the green fluorescence was not seen in the cells treated with doses of TPEN, while brighter in the cells pre-treated with DTDP compared to the untreated group. Moreover, the fluorescence intensity was significantly improved when the cells were pre-treated with Zn^{2+} . The above intracellular imaging results demonstrated that endogenous Zn^{2+} could achieve intracellular fluorescence imaging. In addition, since Zn^{2+} was a cancer-related biomarker for breast cancer, MCF-7 cells exhibit brighter fluorescence than other cells (**Figure S6**). Therefore, the $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe could effectively detect Zn^{2+} in tumor cells through cell fluorescence images, and play a diagnostic function for distinguishing cancer cells from normal cells.

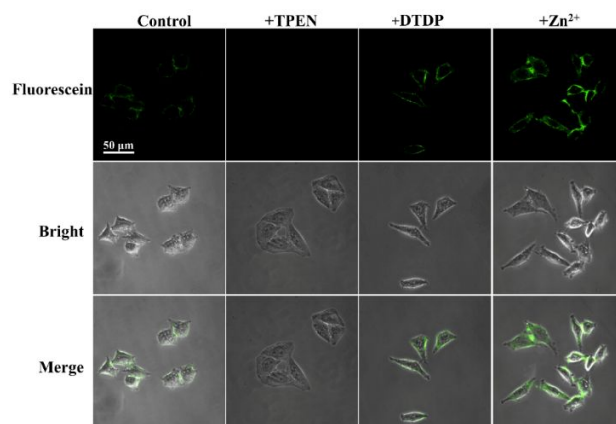


Figure 4. Fluorescence imaging of MCF-7 cells pre-treated with different treatment after incubated with 60 $\mu\text{g mL}^{-1}$ $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe for 4 h.

In Vivo Imaging. Accurate diagnosis can provide effective guidance for cancer treatment. Encouraged by the application of sensing the endogenous Zn^{2+} with $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe in cells, the feasibility of the *in vivo* imaging guided antitumor PTT and PDT was demonstrated in MCF-7 tumor-bearing nude mice. Firstly, after intratumorally injected with $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe (**Figure 5A**), fluorescence intensity of FAM in the nanomachine amplification system increased gradually and reached a maximum at 6 h, which indicating that our $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe could be used for sensitively detection of Zn^{2+} *in vivo*. Secondly, to further evaluate the feasibility of $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe for MR imaging, the MRI signals of $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe dispersions were first observed. As displayed in **Figure 5B**, T_2 -weighted MR imaging shows a concentration-dependent darkening effect. Such a high r_2 relaxivity (179.3 $\text{mM}^{-1}\cdot\text{s}^{-1}$) indicated that $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ has great potential as a T_2 contrast agent. Encouraged by prosthesis data, the MR imaging of the $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe was then observed *in vivo*. Compared to PBS group, the MRI intensity was visibly increased in the tumor position, verifying that the nanoprobe could aggregate in the tumor site.

Considering that $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs have excellent photothermal effect *in vitro*, the feasibility of employing $\text{Fe}_3\text{O}_4@\text{PDA}$ as a photothermal agent *in vivo* was further evaluated. Compared with the PBS group and the $\text{Fe}_3\text{O}_4@\text{PDA}$ group under no external magnetic field, the $\text{Fe}_3\text{O}_4@\text{PDA}$ group under external magnetic field achieved a

more efficient photothermal effect, and the temperature of the tumor region enhanced, reaching 53.1 °C at the tumor center (Figure 5C). These results suggested that a photothermal effect could be generated by the Fe₃O₄@PDA NPs, which strengthened with magnetic targeting depending on the aggregation of more Fe₃O₄@PDA NPs.

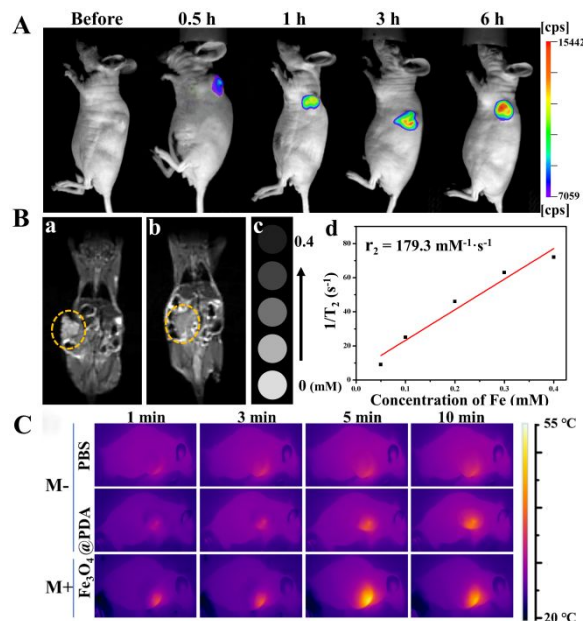


Figure 5. (A) Time-dependent *in vivo* fluorescence images after intratumoral injection of Fe₃O₄@PDA@DNA nanoprobes solution in xenografted MCF-7 tumors. (B) T₂-weighted MRI of mice after intratumoral injection of PBS (a) and Fe₃O₄@PDA@DNA nanoprobes solution (b). The orange circles represent the tumor regions. T₂-weighted MR imaging corresponding to (c) and T₂ relaxation rates at various concentrations of Fe₃O₄@PDA@DNA nanoprobes (d). (C) *In vivo* thermographic imaging after intratumorally injected with PBS, Fe₃O₄@PDA@DNA nanoprobes solution irradiated by 808 nm laser with or without the guidance of magnetic field.

Photodynamic and Photothermal therapy. For multimodal therapy, PTT and PDT provided by the Fe₃O₄@PDA@DNA nanoprobes was performed on MCF-7 cells. The calcein-AM/propidium iodide (PI) cell-survival assay could visually observe the cell viability and simply distinguish between living cells and dead cells. As displayed in Figure S8A, the group of multimodal therapy in cells incubated with the Fe₃O₄@PDA@DNA nanoprobes and exposed to 808 nm laser and white light jointly shows great cytotoxicity, and almost no living cells were seen. In contrast, the group PTT and the group of PDT only showed certain cytotoxicity, and each of them failed to achieve the desired therapeutic effect. Afterward, apoptosis rate of MCF-7 cells detected by flow cytometry was used to further assess treatment efficiency. As displayed in Figure S8B, MCF-7 cells under PDT and PTT showed higher apoptosis (63.4% of apoptotic cells). As expected, lower apoptosis was detected in the group PTT (20.7%) and the group of PDT (36.1%). In addition, CCK-8 cell viability assays were similar to flow cytometry assays. After MCF-7 cells incubated with the Fe₃O₄@PDA@DNA nanoprobes (60 μg mL⁻¹, 2.8 μM), the cell viability was 69.3% under PTT and 58.1% under PDT, but an excellent 34.5% for PTT and PDT (Figure S9). Those above results show that the

remarkable therapeutic efficacy of the Fe₃O₄@PDA@DNA nanoprobes was achieved by excellent PDT efficacy and PTT efficacy at the cellular level.

Encouraged by excellent experimental results mentioned above, the *in vivo* photothermal-photodynamic dual-modal anticancer capability of Fe₃O₄@PDA@DNA nanoprobes was investigated. Tumor changes in each group of mice were shown in Figure 6A. Compared to slight tumor growth inhibition of the control group, the PDT & PTT group displayed dramatic tumor growth inhibition. In addition, the PTT group and the PDT group also showed certain tumor growth inhibition. From the digital photos of H&E staining of tumor sections, it was apparent that the tumor cells exhibited serious cell damage due to large area of cell necrosis and apoptosis after the combined PTT and PDT. The data in Figure 6B further indicated that the PDT&PTT group dramatically inhibited the tumor growth compared to other group after 16 days. We also carried out the histological analysis on the main organs after 16 days of the first treatment. The results confirmed that the main organs of four groups all exhibit no pathological changes and no abnormal phenomenon (Figure S10). Moreover, blood biochemical parameters of liver and kidney function were analyzed at 1 day, 8 days and 16 days after the synthetic PTT and PDT (Figure S11). No significant difference was observed on each indicator over time, which demonstrated the safety of Fe₃O₄@PDA@DNA nanoprobes during the treatments. In these treatments, the body weight of the mice exhibited only slight fluctuation (Figure 6C), further suggesting negligible side effects were produced during the treatment. The above-mentioned results revealed that the Fe₃O₄@PDA@DNA nanoprobes could exhibit high tumor inhibition rate and excellent biocompatibility.

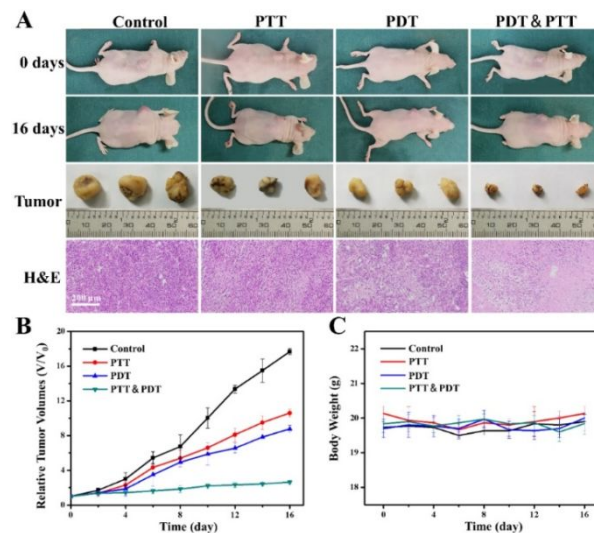


Figure 6. (A) Photographs of differently treated mice, excised MCF-7 tumors and H&E staining of tumor sections (16th d of the observation period). (B) Relative tumor volumes growth curves. (C) Relative MCF-7 tumor-bearing nude mice weights vary curves.

CONCLUSIONS

In summary, a novel multifunctional Fe₃O₄@PDA@DNA nanoprobes based on DNA molecular machine triggered by endogenous ATP to carried out repetitive TSDR was developed to detect Zn²⁺ levels and guide synergistic

photothermal therapy and photodynamic therapy. It had been proved that the as-fabricated theranostic nanoprobe displayed superior detecting sensitivity of Zn^{2+} . The experimental results suggested that the designed nanoprobe could efficiently recognize tumor cells and normal cells based on the detection of intracellular Zn^{2+} levels. Through the optimization of nanostructures, the $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe possessed excellent superparamagnetism, molar extinction coefficient, and photothermal performance, which promote the perfect applications in MRI and PTT. Furthermore, efficient PDT was also achieved by much $^1\text{O}_2$ produced from $\text{Fe}_3\text{O}_4@\text{PDA}@\text{DNA}$ nanoprobe under light irradiation. All in all, this work demonstrated that a unique understanding of the design of theranostic nanoprobe achieved dual-activatable fluorescence/MRI bimodal strategy in cancer imaging and simultaneously efficient tumor treatment.

ASSOCIATED CONTENT

Supporting Information

The experimental procedure contain: Experimental sections, Results and discussion, High resolution XPS of $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs (Figure S1), Photothermal property of $\text{Fe}_3\text{O}_4@\text{PDA}$ NPs (Figure S2), Fluorescence spectra of condition optimization (Figure S3), Photograph of Fe_3O_4 NPs of magnetic property and stability (Figure S4), Time optimization (Figure S5), Fluorescence imaging of different cells (Figure S6), Detection of intracellular $^1\text{O}_2$ (Figure S7), Cell survival and apoptosis assays (Figure S8), Cytotoxicity of the nanoprobe (Figure S9), H&E stained tissue sections (Figure S10) and Blood biochemical analysis (Figure S11). Supplementary data regarding this article are available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENT

This work was supported by the National Natural Science Foundation of China (21565002), Natural Science Foundation of Jiangsu Province (BK20171174).

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